

23 Chelton Drive, L4E 4A9, Richmond Hill ON | (437)-557-6980 | mstroiki@uoguelph.caLinkedIn: [linkedin.com/in/maksim-stroikin](https://www.linkedin.com/in/maksim-stroikin) | Portfolio: maksimstroikin.github.io/portfolio

Automated Pump Transfer Conveyor (Petra Hygienic)

*CLICK PLUS PLC, C-more HMI, Custom Wiring, Pneumatics***August 2026**

What?

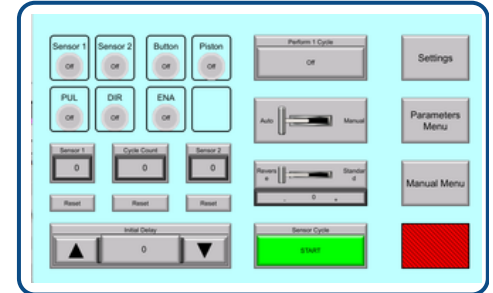
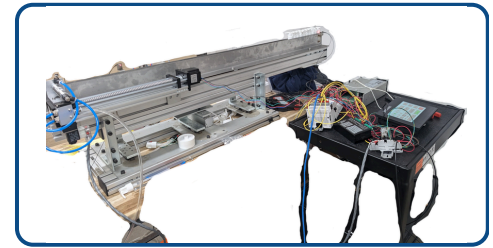
Designed and built a 300mm linear ball-screw actuator system to act as an improved replacement for an \$800k+ dual-arm robot system, transporting various caps and pumps

How?

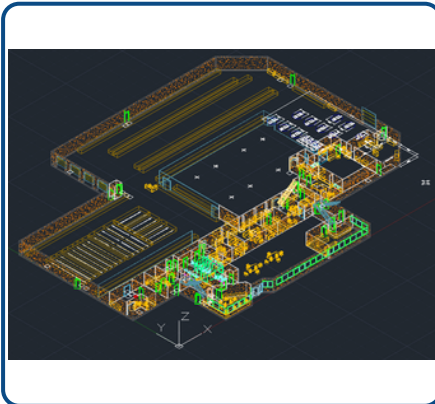
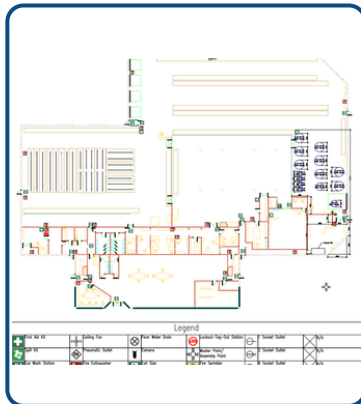
Performed all custom electrical wiring and programmed CLICK PLC ladder logic alongside C-more HMI controls, integrating a stepper driver, relay, 2 fiber-optic sensors, and a 5 port pneumatic solenoid valve.

Results

Achieved full operational functionality with multi-variable HMI parameter control (speed, acceleration, stroke distance, and operating modes), engineered to interface seamlessly with existing plant PLCs and sensor arrays.



Industrial Facility Layout & 3D Modeling (Petra Hygienic)

April - August 2026*CLICK PLUS PLC, C-more HMI, Custom Wiring, Pneumatics*

What?

Generated a complete 3D digital model and 2D architectural layout of an active 120,000 sq. ft., two-story manufacturing facility to replace fragmented legacy files.

How?

Remeasured the physical building geometry and modeled over 75 pieces of equipment, 10 active production lines, and 52 functional zones across custom perspective and isometric views.

Results

Delivered a versatile, 1:1 scale, 3D facility model actively used across departments for audit purpose, new machinery placement planning, and establishing clear emergency evacuation routes.

STM32 Arm-Motor Drive PCB (UofG Robotics Team)

*KiCad, Circuit Design, CAN-FD, RS485***October 2025**

What?

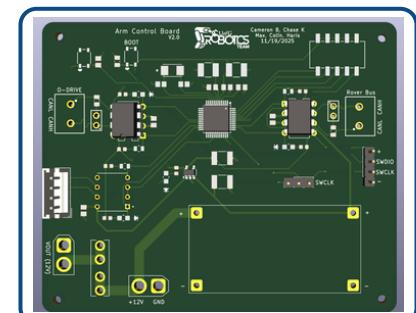
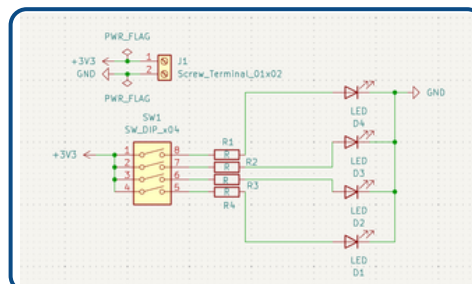
Designed and routed a 4-layer arm-control PCB to interface a rover's STM32 controller, Dynamixel actuators, and ODrive motor controllers via a CAN bus.

How?

Successfully completed the digital layout and schematic integration of CAN transceiver circuitry and a MAX485 interface to convert STM32 UART signals to RS485.

Results

Engineered the on-board power tree with a 12V-to-5V buck converter and 5V-to-3.3V LDO, strategically arranging high-current inputs alongside control buses in KiCad.



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Kinder-Maxi Basketball Launcher (Engineering Design II Course)

*KiCad, Circuit Design, CAN-FD, RS485***January 2025 – April 2025**

What?

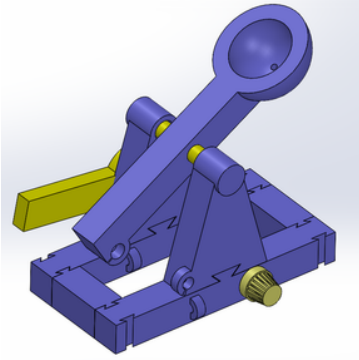
Designed a spring-powered basketball launcher toy utilizing a custom ratchet and pawl mechanism for controlled kinetic energy release.

How?

Created hand-drawn technical sketches, modeled all components in SolidWorks, and sliced/printed parts using a Prusa 3D printer.

Results

Iteratively refined tolerances to ensure the fully disassembled assembly fit inside a 10cm Kinder Maxi Egg while maintaining a 3-meter launch range.



Toy Elephant 3D Digital Assembly (Engineering Design II Course)

January 2025 – April 2025*SolidWorks, 3D Assembly, Reverse Engineering*

What?

Reverse-engineered a physical 12-in-1 solar hydraulic kit to digitally replicate its mechanics

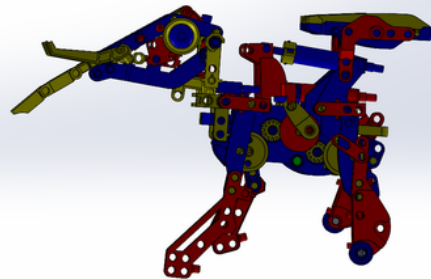
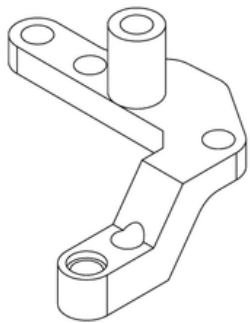
How?

Precision-measured each component using a caliper and 3D modeled 68 unique components within SolidWorks.

Results

Produced a complete digital assembly of 139 total parts, generating 2D technical drawings and an animated exploded view of the model.

Isometric View



STM32 Arm-Motor Drive PCB (UofG Robotics Team)

*Arduino, ESP32, C++, System Integration***September 2024 – November 2024**

What?

Built an interactive pinball machine that automates aluminum recycling by activating gameplay upon detecting a deposited pop-tab.

How?

Integrated an ESP32, 6 sensors, an LCD display, 4W speakers, and 30 LEDs using C++.

Results

Programmed real-time scoring logic and wireless data uploading to a live leaderboard, successfully deploying the machine at a campus carnival.

